Task 19: Create a Git repository for a sample project and demonstrate basic Git operations such as initializing a repository, staging files, committing changes, and maintaining version history. Push the repository to GitHub and manage it using remote repositories.

Aim

To create a **Git repository** for a sample project and demonstrate basic Git operations such as **initializing a repository, staging files, committing changes, and maintaining version history**, and push the project to **GitHub** using remote repositories.

Algorithm

1. Install Git on the system.
2. Create a new project folder.
3. Initialize Git repository using git init.
4. Add files using git add.
5. Commit files using git commit.
6. Create a repository on GitHub.
7. Connect local repo to GitHub using git remote add.
8. Push code using git push.
9. Verify repository on GitHub.
10. Track changes and version history.

Commands Implementation

1. Initialize Repository

git init

2. Add Files

git add .

3. Commit Changes

git commit -m "Initial commit"

4. Connect to GitHub

git remote add origin https://github.com/username/project-name.git

5. Push to GitHub

git branch -M main
git push -u origin main

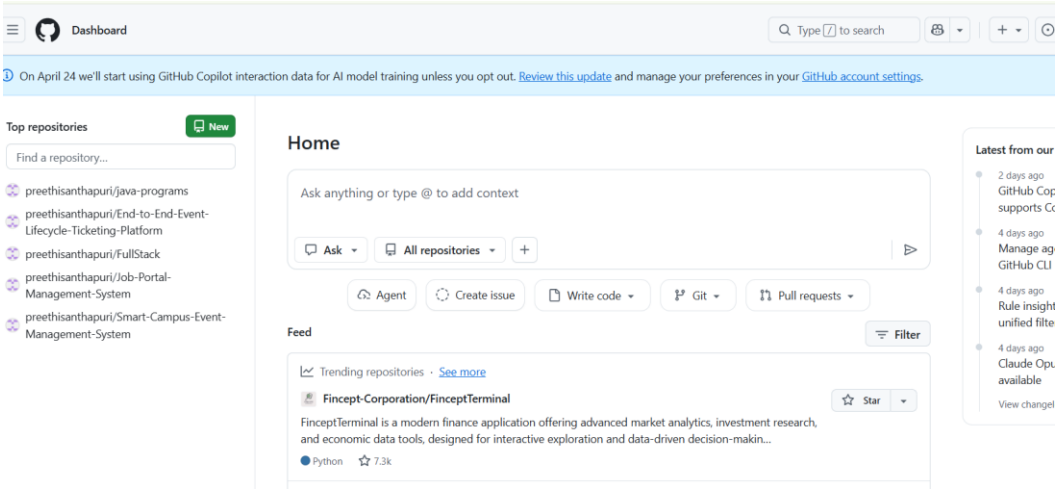
6. Check Status

git status

7. View Commit History

git log

Output:



Result

Git repository successfully created Files staged and committed Connected to GitHub remote repository Code pushed and version history maintained